Class-Incremental SAR Ship Detection and Classification via Context-Robust Exemplar Replay and Multigranularity Knowledge Distillation

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Class-incremental learning based on convolutional neural networks for synthetic aperture radar (SAR) targets has garnered wide attention recently. Unlike incremental SAR target classification, incremental SAR target detection and classification requires the model to detect and classify old targets while learning current ones, which is challenging due to the absence of old targets in the incremental large-scale data. To address the aforementioned issues, this article proposes a class-incremental SAR ship detection and classification method, which combines exemplar replay (ER) and knowledge distillation (KD) to facilitate the learning of the incremental model. Specifically, we propose a context-robust ER method to retain precise scene context between replayed old targets and clutter in the incremental data. By fully leveraging the scene characteristics of the ship targets, a context-robust ER method combines a local context-aware strategy with sea–land segmentation to pinpoint regions with the highest likelihood of being sea areas. Then, it replays exemplars within these regions, effectively reducing context-bias issues. In addition, a multigranularity KD method is introduced, which transfers the knowledge learned by the old detector to the current detector progressively. The multigranularity KD method combines an object mask strategy with spatial-channel attention mechanisms to constrain the current detector to focus on the most important information from the old detector. Experiments conducted on the SRSDD-v1.0 dataset indicate that our method achieves satisfactory performance in incremental detection and classification of SAR ships.

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I. INTRODUCTION

Synthetic aperture radar (SAR) offers continuous observation and imaging capabilities, independent of weather conditions or time of day. With the continuous advancements in SAR systems, high-quality SAR images are increasingly accessible. This growing availability to SAR data has significantly advanced the field of SAR image interpretation. Among them, SAR ship detection and classification is noteworthy, as it plays a crucial role in maritime monitoring. In recent years, benefiting from powerful feature representation capabilities, deep-learning-based methods have demonstrated significant performance advantages over traditional methods [1], [2], [3].

In real-world environments, researchers can incrementally acquire new SAR data, which contain categories that are not encountered previously [4]. To enable the model to detect and classify both new and old targets, a straightforward approach is to retain all old data and retrain the model with both old and new data. However, this approach incurs high storage costs and time consumption [5]. Although fine-tuning the old detector with new data can significantly reduce storage costs and runtime, it comes with a drawback. As shown in Fig. 1, the new detector tends to focus on learning the new data during fine-tuning, which can overwrite the knowledge previously learned. Therefore, the performance on old targets declines, which is known as catastrophic forgetting [6]. Incremental learning offers a promising paradigm, which enables models to learn new classes from new data while retaining previously learned knowledge.

Incremental learning methods can be divided into two approaches: regularization-based and replay-based methods. The former typically employs regularization constraints to alleviate forgetting. Learning without forgetting (LwF) [7], [8] leveraged knowledge distillation (KD) to align the output responses of the teacher and student models to alleviate forgetting. Li et al. [9] proposed the class-incremental learning based on mutual information maximization (CIL-MMI) method for incremental SAR target classification. By using a maximum mutual information constraint, it avoids the distribution overlap inherent in SAR images. Wang et al. [10] improved discriminability between SAR target classes in the embedding space by incorporating a regularization term that combines Euclidean and cosine distances. Yu et al. [11] proposed an incremental SAR target classification method, which combines the feature- and response-level KD to transfer the learned knowledge. The core of replay-based methods is to store a small number of representative old targets, which are used to update the model in the incremental training stage. The replay method was first proposed by incremental classifier and

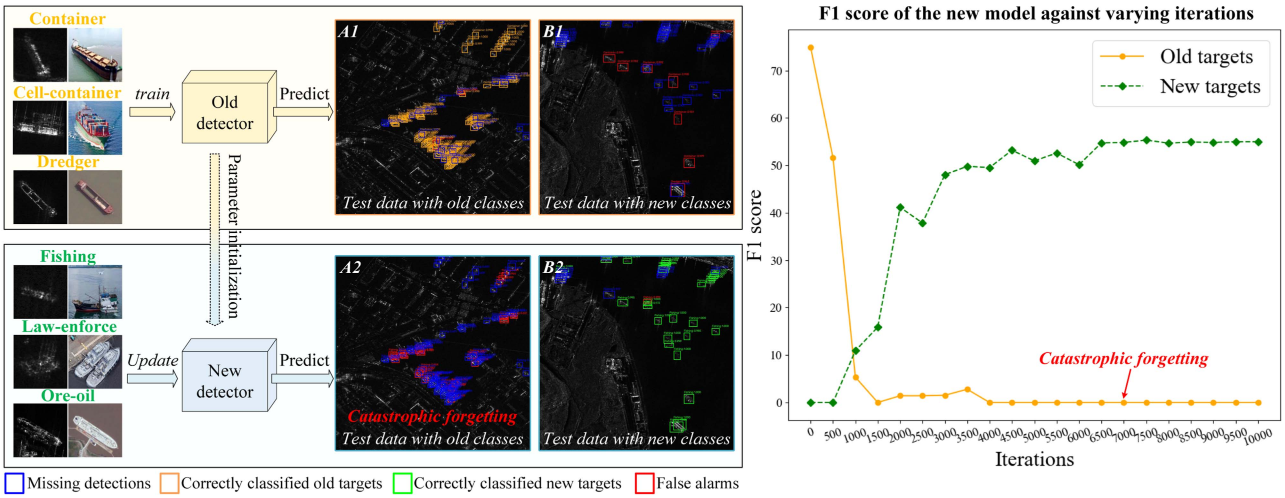

Fig. 1. Diagram of *catastrophic forgetting* issues. In the incremental training stage, the absence of annotations for old targets in the newly introduced data leads the detector to focus predominantly on accurately learning new targets rather than preserving knowledge of old targets. This often results in significant updates to the model parameters, which can overwrite previously learned historical data. As a consequence, the updated detector (new) exhibits a marked decline in detection and classification performance for old targets, as illustrated in subfigure "A2," a phenomenon referred to as catastrophic forgetting.

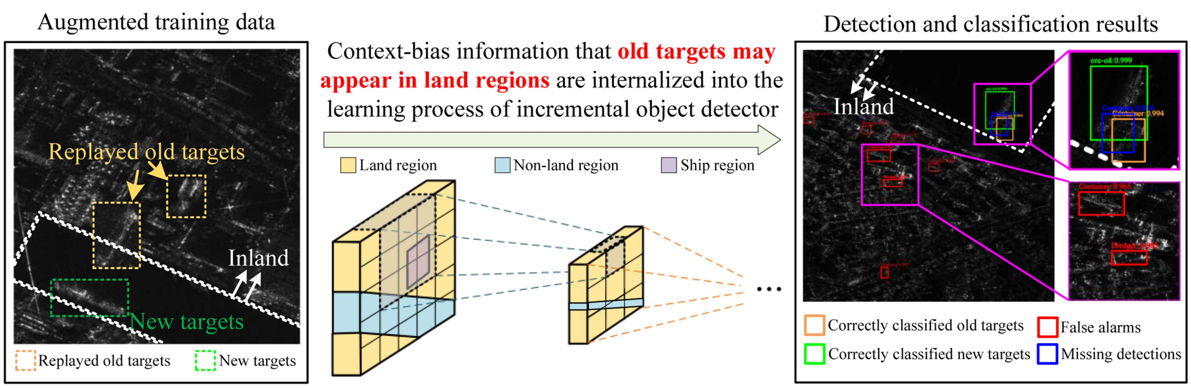

Fig. 2. Diagram of *context-bias* issues. The context-bias issues, stemming from the potential appearance of old targets in land regions, can misguide the learning process of the incremental object detector, ultimately impacting the accuracy of ship target detection.

representation learning (iCaRL) [12], which selects a few old targets nearing the class centers. Inspired by iCaRL, many incremental SAR target classification methods further explored the exemplar selection. Some methods focused on selecting discriminative exemplars. For instance, Li et al. [13] adopted a class-center exemplar selection method, while Dang et al. [14] adopted a decision-boundary exemplar selection method. Other methods [15], [16] focused on selecting diverse exemplars. The replay-based methods sacrifice limited storage space to achieve significant performance improvements and have gradually become a research hot spot in incremental learning.

Recent studies have also explored incremental object detection and classification. Shmelkov et al. [17] pioneered the introduction of LwF [7] into incremental object detection and classification. It leverages KD based on the Fast region-based convolutional neural network (R-CNN) [18]. Since the performance and speed of Fast R-CNN are limited, some researchers [4], [19], [20] soon introduced incremental learning to Faster R-CNN [21] or fully convolutional one-stage object detection (FCOS) [22]. In addition, several works [23], [24], [25] applied exemplar replay (ER) methods to incremental object detection and classification methods. They typically selected a small number of representative old targets and randomly replayed them in the incremental data. However, such ER methods inevitably introduce severe context-bias issues. The context-bias information may mislead the learning process of the incremental object detector, resulting in the potential for false alarms on land clutter and missing detections of real ships as shown in Fig. 2. Building on existing replay-based incremental learning methods, this article further explores how to utilize the scene characteristics of ship targets to mitigate the context-bias issues. Meanwhile, designing KD methods for incremental SAR object detection and classification is also a challenge that needs to be addressed.

1) How to preserve accurate scene context of replayed old targets in augmented training data? This article

identifies the primary reason why replay strategies have not been successfully applied to incremental object detection and classification is the significant context-bias issue inherent in existing ER methods. The context-bias information between clutter and replayed old targets will misguide the learning of the incremental object detector, resulting in higher false alarms and missing detections, as shown in Fig. 2.

2) How to robustly transfer the knowledge learned by the old detector through KD? Unlike incremental SAR object classification methods, incremental SAR object detection and classification methods include both detection and classification tasks. Applying distillation constraints only to the classification heads is insufficient to transfer the knowledge related to old target detection and classification, leading to suboptimal performance.

To address these issues, we propose our solutions in this article. To mitigate context-bias issues that arise when replaying old targets in existing strategies, we propose a context-robust ER method. In large-scale SAR data, ship targets exhibit obvious scene characteristics, which are reflected in the locations of ship targets. In real-world scenarios, ship targets may be situated offshore, nearshore, or on river surfaces, making it impossible for all surrounding vicinity to be entirely composed of land. Combining the aforementioned scene characteristics to improve ER methods is the primary focus of our research. The context-robust ER method constructs a segmented circle centered on the new targets in the incremental data and evaluates the sea probability in different regions by combining sea–land segmentation techniques. We infer that the region with the highest sea probability is the optimal region for replaying exemplars. The context-robust ER method significantly alleviates the context-bias issues inherent in traditional ER methods, enabling the detector to better learn to discriminate between ship targets and clutter. In addition, a multigranularity KD method is proposed to progressively transfer knowledge learned by the old detector. The multigranularity KD method adds distillation constraints at different levels between the new and old detectors, which includes scene-, instance- and response-level distillation losses, effectively ensuring the performance of knowledge transfer. Technically, our KD method combines object masks and attention mechanisms to constrain the current detector to focus on the most important knowledge from the old detector.

Our main contributions can be summarized as follows.

1) *The context-robust ER method is proposed to preserve accurate scene contexts between replayed targets and clutter in the incremental training stage.* By fully utilizing the distinct scene characteristics of ship targets, our replay method alleviates the context-bias issues inherent in traditional replay methods, thereby enhancing the ability of the detector to accurately perceive the relationship between ship targets and clutter. To our knowledge, we are the first to incorporate scene characteristics of ship targets into incremental SAR ship detection and classification.

2) *Multigranularity KD is proposed to transfer the knowledge learned by the previous detector to the current detector progressively.* By incorporating object localization information and attention mechanisms, our distillation method constrains the current detector to focus on the most important information from the previous detector in a progressive KD paradigm, which enhances the overall performance of KD.

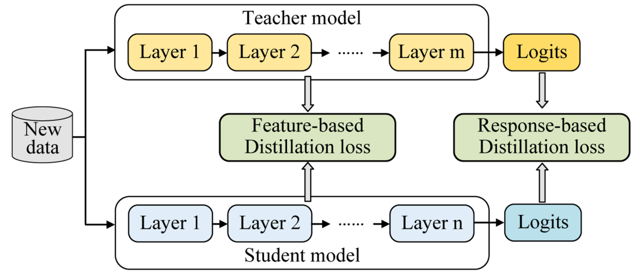

Fig. 3. Diagram of the teacher–student framework for KD.

The rest of this article is organized as follows. Section II covers preliminary knowledge. Section III details the proposed method. Section IV presents experimental results and analysis. Finally, Section V concludes this article.

II. PRELIMINARY

To facilitate a better understanding of our method, this section provides an overview of the basic concepts of KD and land–sea segmentation.

A. Knowledge Distillation

Recent regularization-based incremental methods have evolved into a teacher–student framework [7], [8], [26], as illustrated in Fig. 3. This approach is founded on the observation that deep models exhibit robust representation learning capabilities [27], adept at acquiring multilevel feature representations. Consequently, the outputs from the teacher model can be utilized to guide the training of the student model. The primary goal is to either replicate the teacher's final prediction or to align feature activations between the teacher and student models. The distillation loss for response- and feature-based knowledge transfer is formulated as follows:

$$\mathcal{L}_{\text{distill}} = \mathcal{L}^{\text{fea}}(\Phi_t(f_t(x)),\quad \Phi_s(f_s(x))) + \mathcal{L}^{\text{res}}(z_t, z_s) \quad (1)$$

where $\mathcal{L}^{\text{fea}}$ and $\mathcal{L}^{\text{res}}$ denote the similarity function for aligning the feature maps and the output logits, respectively, transformation functions $\Phi_*(\cdot)$ are applied to map feature maps $f_t(x)$ and $f_s(x)$ to a new unified embedding space, and z_t and z_s denote the output logits of the teacher and the student, respectively.

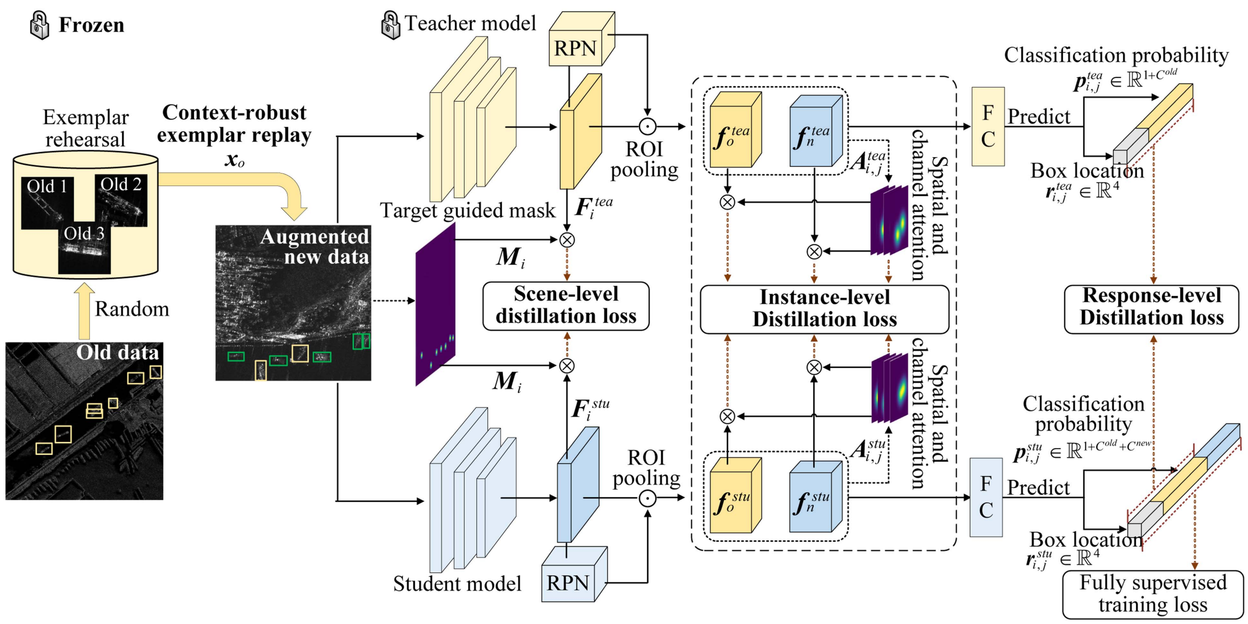

Fig. 4. Overview of the proposed method. Our method includes two innovative modules: 1) the context-robust ER module, where old targets are replayed in the correct scene contexts in the incremental data; and 2) the multigranularity KD module, which comprehensively transfers the previous knowledge to the current model.

B. Sea–Land Segmentation

Sea–land segmentation is an image segmentation technique that differentiates the sea surface from land in an image. Considering that unsupervised sea–land segmentation techniques do not rely on large-scale pixel-level annotated data and deep-learning-based segmentation models suffer from catastrophic forgetting [28], this article adopts a simple and fast sea–land segmentation method Otsu [29]. Otsu determines the optimal threshold T^* for image binarization by maximizing the between-class variance, which quantifies the separability of foreground and background pixel intensities. The method computes the normalized histogram of the grayscale image to represent the probability distribution of pixel intensities. Based on a given threshold T, the image is divided into two classes: foreground and background. The optimal threshold T^* is obtained by maximizing the between-class variance

$$\sigma_b^2(T) = \omega_1(T) \cdot \omega_2(T) \cdot (\mu_1(T) - \mu_2(T))^2 \tag{2}$$

where $\omega_1(T)$ and $\omega_2(T)$ are probabilities of the foreground and background classes, respectively, and $\mu_1(T)$ and $\mu_2(T)$ denote their mean intensities. Otsu evaluates $\sigma_b^2(T)$ across all possible thresholds and selects T^* as the one that maximizes this value.

III. PROPOSED METHOD

A. Overall Framework

As shown in Fig. 4, the proposed method adopts a teacher–student dual model framework for incremental SAR ship detection and classification. The teacher model is a detector well trained on old data. The exemplar rehearsal is constructed in the training stage of the teacher model, which randomly stores a limited number of old targets. In the incremental training stage of the student model, the parameters of the teacher model are frozen, and the context-robust ER and multigranularity KD methods are both employed to guide the learning of the student model. Technically, upon the arrival of incremental data $D_{\text{ind}} = \{X_i, Y_i\}_{i=1,\dots,N_{\text{ind}}}$, where X_i represents the ith training images, Y_i consists of corresponding ground-truth bounding box coordinates $b_{ij} \in \mathbb{R}^4$ and class labels $c_{ij} \in \mathbb{C}^{\text{current}}$, where $\mathbb{C}^{\text{current}}$ is the number of target classes in the current task, and N_{ind} is the number of training images in the current task. The context-robust ER method selects old targets from the exemplar rehearsal, and then, it replays them in the accurate scene contexts in the new data, resulting in augmented new data $\{\tilde{X}_i, \tilde{Y}_i\}_{i=1,\dots,N_{\text{ind}}}$. The augmented new data are fed into both the student model and the teacher model for incremental training. On the one hand, the augmented new data directly steer the parameter optimization of the student model, yielding fully supervised training loss $\mathcal{L}_{\text{R-CNN}}$. On the other hand, through the multigranularity KD method, the knowledge learned by the teacher model is comprehensively transferred to the student model, generating multigranularity KD losses $\mathcal{L}_{\text{distill}}$. Through the combination of fully supervised training loss and multigranularity KD losses, the proposed method effectively retains the previously learned detection and classification knowledge for old targets while learning to detect and classify new targets. The incremental training loss can be given as follows:

$$\mathcal{L} = \mathcal{L}_{\text{R-CNN}} + \lambda \cdot \mathcal{L}_{\text{distill}} \tag{3}$$

where $\mathcal{L}_{\text{R-CNN}}$ and $\mathcal{L}_{\text{distill}}$ represent the fully supervised training loss for the current detector and KD, respectively, and λ represents the training parameter (in our experiments $\lambda = 1.0$). We will provide a detailed introduction of the

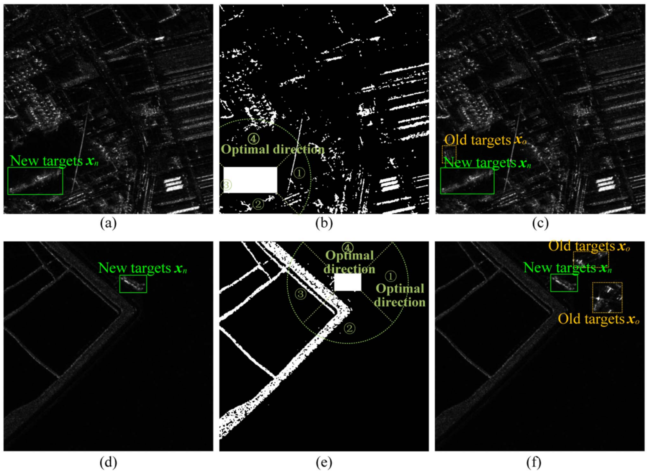

Fig. 5. (a)–(f) Diagram of the context-robust ER method. The first column presents two new training images. The second column presents the intermediate results of the context-robust ER method on the incremental data. The third column presents the augmented new data after replaying stored old targets.

proposed context-robust ER and multigranularity KD in the subsequent sections.

B. Context-Robust ER

Due to extremely limited storage space, the ER method only stores a limited amount of old data, which is replayed in the incremental training stage. Unlike classification tasks, training data in the incremental object detection and classification tasks consist of large-scale images, and directly storing these large-scale images undoubtedly incurs extremely high storage costs. Meanwhile, the new large-scale images also contain clutter information similar to that in the old training images. Therefore, this article stores only old targets rather than the original large-scale images. Following this, the old targets will be replayed in the new data in the incremental training stage of the student model.

1) *Storing Old Target Slices From the Old Data:* Our exemplar selection method is to select proposals with high localization accuracy and correct classification results. In particular, old data are fed into the well-trained old model, generating a large number of proposals. Then, we compare the interaction over union (IoU) between the proposals and the ground-truth labels and sort them in descending order of IoU. After removing misclassified predictions, we obtain the exemplars. Following mainstream replay-based methods [30], we select exemplars with the highest localization accuracy for each class, ensuring that they make up $p\%$ of the total exemplars in that class (in our experiments, $p = 10$).

2) *Replaying Old Target Slices in the New Data:* In the incremental training stage of the student model, we adopt the context-robust ER method, which combines the local context-aware technique with a sea–land segmentation, as shown in Fig. 5, which fully utilizes the scene characteristics that ship targets are adjacent to sea surface regions.

When replaying old target slices in new data, our context-robust ER method needs to determine the number of exemplars assigned to each image. Following the settings in [25], we ensure 1:2 ratio between replayed exemplars and new targets. In detail, we calculate the average number of new targets per image in the new data as u and further set the maximum number of replayed exemplars per image as $\lfloor u/2 \rfloor$, where $\lfloor \cdot \rfloor$ denotes the floor function. Given a new image X_i, we randomly point a new target x_n from ground-truth labels and an old target x_o from the exemplar set. During a single ER process, we perform data augmentation on the old target x_o, including Gaussian noise, horizontal flipping, or rotation operations. To avoid overlap between targets, we mask existing targets based on the annotations. Then, we draw a segmented circle at the centroid of the new target x_n with a radius of r (in our experiments, $r = 50$) and divide it into four quadrants, as shown in the second column in Fig. 5. Following this, we assess the sea probability for each of the four quadrants. We apply a sea–land segmentation algorithm to the SAR images to obtain the corresponding binary segmentation results. The sea probability is calculated as the proportion of sea pixels (black pixels) to the total number of pixels in the current quadrant. Based on the previous analysis, the quadrant with the highest sea probability is the most likely to be sea surface regions. We use a replay box with the same resolution as the old target x_o to randomly sample n times (in our experiments, $n = 25$) within the optimal quadrant, and then, we replay the old target x_o in the replay box with the highest sea probability. The aforementioned replay process is performed $\lfloor u/2 \rfloor$ times. Note that if the highest sea probability is below preset threshold 0.90, we consider that replaying the current old target may introduce context-bias issues and thus skip the old target. The number of old targets replayed in an image may not always reach $\lfloor u/2 \rfloor$. Finally, we can obtain the augmented training data $\{\tilde{X}_i, \tilde{Y}_i\}$ corresponding to the training data $\{X_i, Y_i\}$.

Algorithm 1 shows the pseudocode of the context-robust ER method.

C. Multigranularity KD

KD centers on the teacher–student training paradigm, where a well-trained teacher model guides the learning of the student model, transferring its knowledge. In incremental object detection and classification, the ability to accurately detect and classify old targets is referred to as *knowledge*. KD transfers insights from the teacher to the student model, reducing catastrophic forgetting. Traditional KD methods are primarily designed for object classification tasks, while object detection and classification is a combination of detection and classification tasks. The transfer of only classification knowledge is insufficient. Therefore, this article designs the multigranularity KD method, as shown in Fig. 4, incorporating scene-level feature distillation, instance-level feature distillation, and top-level response distillation to mitigate catastrophic forgetting, offering the student model comprehensive and diverse abstractions.

1) *Scene-Level Feature Distillation:* For most deep-learning-based two-stage object detection models, the entire model can be divided into three parts: the backbone

Algorithm 1: Pseudocode of the Context-Robust ER Method.

```
Require: Training data {X_i, Y_i}, sea–land segmentation technique seg(·), training parameters r and n, parameter u;
Ensure: Augmented training data {X̃_i, Ỹ_i};
for each X_i, Y_i in training data {X, Y} do
    for k = 1 to ⌊u/2⌋ do
        Obtain new target x_n sampled from the training image X_i according to the annotations Y_i;
        Obtain old target x_o, sampled from the exemplar set, and perform data augmentation on it;
        Divide the areas around the new target x_n into quadrants based on parameter r and compute the sea probability in different quadrants using segmentation result seg(X_i^quadrant);
        Randomly sample n times within the optimal quadrant and replay the old target x_o in the replay box with the highest sea probability, provided that it is below the threshold of 0.90;
    end for
end for
Obtain the augmented data {X̃_i, Ỹ_i}.
```

network for feature extraction, the region proposal network (RPN) for generating region proposals, and the region of interests (RoI) head for classification and regression. To retain the backbone network's understanding of the local context between ship targets and clutter, we propose target-guided scene-level feature KD. In detail, the core idea is to apply an expansion mask on the feature maps based on the target annotations, constraining the student model to focus on the local context of the targets rather than the entire feature maps. We begin by initializing a mask $M_i \in \mathbb{R}^{C\times H}$ with all zeros for the augmented data $\{\tilde{X}_i, \tilde{Y}_i\}$. Next, to enhance the detector's understanding of the context around each target, we expand each target bounding box b_{ij} in width and height by a factor of q (in our experiments, $q = 1.2$). This adjustment includes additional scene information. Finally, we set the elements of M_i corresponding to the expanded bounding boxes in the image to one. The augmented new data are fed into both the student model and the well-trained teacher model, resulting in scene-level feature maps F_i^{stu} and F_i^{tea} after passing through the corresponding backbone networks. We choose ResNet-50 with feature pyramid network (FPN) as our backbone network; therefore, it will output scene-level feature maps at different resolutions $F_i^{\mathrm{stu}} = [F_{i1}^{\mathrm{stu}}, .., F_{ir}^{\mathrm{stu}}]$ and $F_i^{\mathrm{tea}} = [F_{i1}^{\mathrm{tea}}, .., F_{ir}^{\mathrm{tea}}]$, where r represents the number of residual blocks with a default of 4. The resolutions of the scene-level feature maps are sequentially 1/4, 1/8, 1/16, and 1/32 of the original SAR image size. After passing through multiple residual blocks, the relative position of the target in the image remains unchanged. Therefore, the expansion mask M_i^{stu} and M_i^{tea} are downsampled to obtain masks $[M_{i1}^{\mathrm{stu}}, .., M_{ir}^{\mathrm{stu}}]$ and $[M_{i1}^{\mathrm{tea}}, .., M_{ir}^{\mathrm{tea}}]$ at corresponding resolutions. The scene-level feature distillation loss is defined as follows:

$$\mathcal{L}_{\mathrm{scene}} = \frac{1}{N\cdot R}\cdot\sum_{i=1}^{N}\sum_{r=1}^{R}||F_{ir}^{\mathrm{stu}}\times M_{ir}^{\mathrm{stu}} - F_{ir}^{\mathrm{tea}}\times M_{ir}^{\mathrm{tea}}||_2^2 \tag{4}$$

where N represents the number of training datasets and R equals 4. Compared to constraining the entire feature maps between the teacher model and the student model, our scene-level feature distillation mitigates the negative impact of redundant clutter information, helping the incremental model focus on discriminative information around the targets.

2) *Instance-Level Feature Distillation:* The RPN is responsible for discriminating suspected targets from clutter, generating region proposals that are likely to contain targets for subsequent classification and regression. A limitation of existing RPN distillation is its neglect of features within individual proposals, leading to the distilled model overlooking crucial information and resulting in suboptimal performance [25].

Therefore, we propose the attention mechanism-guided instance-level feature distillation loss, which adaptively focuses on the most important information by aligning the spatial-channel attention maps and masking the features of each proposal. First, we need to select appropriate RoI features for instance-level feature distillation. Existing methods [25] typically input new data into the teacher model, randomly selecting 64 proposals $f_{i,j}^{\mathrm{tea}}$, $j = 1, \ldots, 64$, from the top 256 confidence-ranked proposals generated by the RPN, and then choosing the corresponding proposals $f_{i,j}^{\mathrm{stu}}$, $j = 1, \ldots, 64$, from the student model. The selected proposals are then processed through RoI pooling to obtain RoI features, which are used for KD. However, the number of targets is not fixed, and setting an excessively large number of proposals may include many false alarms. Using false alarms for KD is likely to negatively impact performance of the KD. Therefore, this article adopts a dynamic proposal selection strategy: we retain only the proposals $f_{i,j}^{\mathrm{tea}}$ with confidence scores greater than 0.7, instead of consistently selecting the top-scoring proposals from each image. Next, to focus on the most informative parts of the RoI features, we calculate the spatial and channel attention maps $A_{i,j}^{\mathrm{tea}} = [A_{i,j}^{\mathrm{tea},s}, A_{i,j}^{\mathrm{tea},c}]$ for each $f_{i,j}^{\mathrm{tea}} \in \mathbb{R}^{c\times h\times w}$. Inspired by Zagoruyko and Komodakis [31], the spatial attention maps $A_{i,j}^{\mathrm{tea},s} \in \mathbb{R}^{1\times h\times w}$ are obtained by raising the absolute value of each channel to a power p (in the experiments, $p = 2$) and summing them up. The channel attention maps $A_{i,j}^{\mathrm{tea},c} \in \mathbb{R}^{c\times 1\times 1}$ are obtained by applying global adaptive pooling to each feature channel. The instance-level distillation loss is defined as

$$\mathcal{L}_{\mathrm{instance}} = \frac{1}{N}\cdot\frac{1}{P_i}\cdot\sum_{i=1}^{N}\sum_{j=1}^{P_i}(||A_{i,j}^{\mathrm{tea}} - A_{i,j}^{\mathrm{stu}}|| + ||f_{i,j}^{\mathrm{tea}} - f_{i,j}^{\mathrm{stu}}||\times A_{i,j}^{\mathrm{tea}}) \tag{5}$$

where P_i represents the number of proposals for the ith SAR image. The key differences between the instance-level feature distillation loss and existing distillation methods are that, on the one hand, we implicitly distill the relative location information encoded in the attention maps. On the other hand, by applying the attention maps as masks to the original RoI features, we explicitly emphasize the most important features while suppressing the influence of less significant features.

3) *Top-Level Response Distillation:* The RoI features obtained from the RPN are further input into the RoI heads, producing the corresponding classification probability and box regression results. As shown in Fig. 4, the number of classification heads in the student model increase by C^{new} compared to the teacher model $\mathbb{R}^{1+C^{\text{old}}} \to \mathbb{R}^{1+C^{\text{old}}+C^{\text{new}}}$, where C^{old} and C^{new} represent the number of old new target classes, respectively, and 1 represents the background class. To prevent forgetting in the classification heads, we need to align the dimensions of the classification heads between the teacher and student models and then apply the relevant constraints.

Inspired by Cermelli et al. [32], [33], we develop an unbiased classification loss to retain the classification knowledge of old targets by aligning the probabilities of the background and old target classes in the teacher model. This is because the teacher model is only capable of classifying old targets. When new targets arrive, the teacher model classifies them as background. Therefore, we define the response-level distillation loss with background constraints as follows:

$$\mathcal{L}_{\text{response}} = \sum_{i=1}^{N}\sum_{j=1}^{P_i}\frac{1}{N}\cdot\frac{1}{P_i}\begin{cases}A, & c_{i,j}\in\mathbb{C}^{b}\\ B, & c_{i,j}\in\mathbb{C}^{\text{old+new}}\end{cases} \tag{6}$$

$$A = p_{i,j}^{b,\text{tea}}\cdot\log(p_{i,j}^{b,\text{stu}} + \sum_{c\in C^{\text{new}}} p_{i,j}^{c,\text{stu}}) \tag{7}$$

$$B = \sum_{c\in C^{\text{old}}} p_{i,j}^{c,\text{tea}}\cdot\log(p_{i,j}^{c,\text{stu}}) \tag{8}$$

where $p_{i,j}^{b,\text{tea}}$ and $p_{i,j}^{c,\text{tea}}$ are the classification probabilities of the background class and cth target classes for the teacher model, respectively, $p_{i,j}^{b,\text{stu}}$ and $p_{i,j}^{c,\text{stu}}$ are the classification probabilities of the background class and cth target classes for the student model, respectively, and $c_{i,j}$ represents the label of the jth proposal in the ith training data. Since the classification heads of teacher model are not capable of classifying new targets, (8) only constrains the classification probabilities of the old classification heads when the proposals are positive RoI features $c_{i,j}\in\mathbb{C}^{\text{old+new}}$. When the proposals are negative RoI features $c_{i,j}\in\mathbb{C}^{b}$, (7) applies background constraints by adding the classification probabilities of the new target classes $\sum_{c=1}^{C^{\text{new}}} p_{i,j}^{c,\text{stu}}$ output by the student model to the background class probability $p_{i,j}^{b,\text{stu}}$.

IV. EXPERIMENTS

A. Datasets

This article uses the SRSDD-v1.0 dataset [34], which includes multiclass ship detection and classification in our experiments. Fig. 6 shows some training and test data from the dataset, including SAR images of six types of ship targets and their corresponding optical images. For our specific task, we convert the original annotated rotated bounding boxes into horizontal minimum outer rectangular boxes, following the strategy in [35]. It is derived from the Gaofen-3 and includes 666 SAR images with a resolution of 1 m and 1024 × 1024 pixels. The imaging mode is the spotlight. Among 666 images, 532 images are designated for training, and others are reserved for testing [34]. To train the detector, we crop 512 × 512 subimages from the original large-scale images using a sliding window with a 256-pixel overlap, obtaining 1225 training subimages with at least one ship target. Finally, nonmaximum suppression with an IoU threshold of 0.5 is used to remove overlapping bounding boxes.

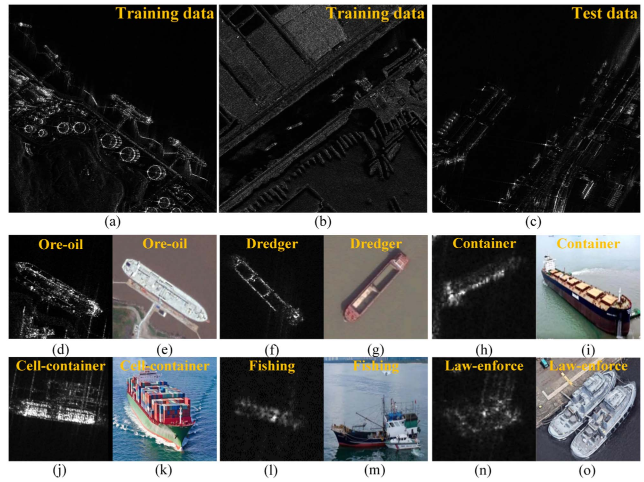

Fig. 6. Several SAR images from the SRSDD dataset. (a)–(c) Some examples of the original large-scale SAR images. (d)–(o) Different types of ship targets in the dataset, including the SAR images and corresponding optical images.

Following the alphabetical order standards, we divide the target classes into different splits, specifically into single-step and two-step increments, as shown in Table I. The annotations are available only for the current target classes, while the previous and future classes are not annotated. It is worth noting that new and old targets may appear in the same training image. To avoid the repetition of the training images across tasks [34], we divide the training images where old and new targets coexist into different splits according to numeric order standards, which ensures practical and reasonable experimental settings.

B. Implementation Details

For a fair comparison, we follow the augmented box replay (ABR) method [25] to use Faster R-CNN with a ResNet-50 as backbone. The training parameters λ, p, r, n, and q are 1.0, 10, 50, 25, and 1.2, respectively. To balance the number of old and new targets, we determine the 1:2 ratio for replayed old targets and new targets following [25]. We train the detector with stochastic gradient descent (SGD), weight decay 10^{-4}, and momentum 0.9. The batch size is 16. The training iterations are 20 000 for each task and the initial

TABLE I
Information Regarding the Category Labels Contained Within Different Tasks

Single-step increments	Task D_0		Task D_1
Target classes	Cell-container, container, dredger		Fishing, law-enforce, ore-oil
Number of subimages	1012		213
Two-step increments	Task D_0	Task D_1	Task D_2
Target classes	Cell-container, container	Dredger, fishing	Law-enforce, ore-oil
Number of subimages	832	231	162

We evenly distribute the training images containing both new and old target classes across different tasks (the sum of training images for different tasks equals the total number of training images), avoiding the repetition of the training images across tasks in mainstream settings.

learning rate is 0.02, with the learning rate decays to one-tenth at iterations 10 000 and 15 000. Our implementation is based on Detectron2 [36].

C. Evaluation Metrics

This article uses macro precision, macro recall, and macro F1 for old targets, new targets, and overall targets to evaluate the detection and classification performance metrics under incremental conditions. The definition is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{9}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{10}$$

where TP represents the number of true positives, FP represents the number of false alarms, and FN represents the number of missing detections. F1 is the harmonic mean of precision and recall.

D. Comparison With Incremental Learning Methods

We conduct experiments on settings with single-step and two-step increments to verify our method. The compared methods can be divided into four parts.

1) *The incremental method with fine-tuning strategies* typically freezes the parameters of the backbone network and RPN and fine-tunes RoI head using incremental data. This method is generally considered the lower bound of incremental learning.
2) *Incremental methods with regularization strategies* [17], [19] typically employ a teacher–student model framework, where the knowledge learned by the teacher model is transferred to the student model through KD constraints.
3) *Incremental methods with replay strategies* [23], [25] store and replay old targets along with the new data to train the detector. Notably, Liu et al. [25] incorporate KD constraints to retain old knowledge learned by the teacher model.
4) *Upper bound* refers to training the model with both new and old training data. It does not suffer from catastrophic forgetting from gradual data introduction and is typically used to evaluate the performance upper bound of incremental learning.

TABLE II
F1 Scores on Settings With Single-Step Increments on the SRSDD-V1.0 Dataset

Method \ F1 (%)	Single-step increments (3+3)		
	Old	New	Overall
Fine-tuning	14.76±1.77	47.39±0.71	31.07±0.95
ILOD [17]	43.34±1.09	49.32±1.88	46.34±0.99
F-ILOD [19]	42.42±1.20	51.77±2.71	47.09±1.36
ORE* [23]	53.21±2.28	57.21±2.80	55.71±0.76
ABR* [25]	<u>60.47</u>±0.81	<u>57.26</u>±1.10	<u>58.87</u>±0.75
Proposed*	**68.61**±0.66	**60.51**±0.62	**64.56**±0.63
Upper bound	75.58±0.22	66.02±0.19	70.52±0.16

We present the mean and standard deviation after three random trials. Bold indicates the best results in each column, while underlined denotes the second best. Methods using exemplars are marked with an asterisk *.

1) *Single-Step Increments:* We conduct experiments on the settings with single-step increments on the SRSDD-v1.0 dataset. As shown in Table II, the F1 score on old targets of the incremental method with fine-tuning strategies is only 14.76%, indicating that this type of method suffers from severe catastrophic forgetting. ILOD [17] and Faster ILOD [19] alleviate the catastrophic forgetting in the incremental training stage to some extent (F1 scores on old targets reach 43.34% and 42.42%, respectively) by transferring the knowledge of detecting and classifying old targets learned by the teacher model to the student model through KD constraints. Meanwhile, as the detected old targets are correctly classified as the corresponding target classes, F1 scores on new targets improves by 1.93% and 4.38% with the ILOD [17] and Faster ILOD [19] methods, respectively. Open world object detector (ORE) [23] employs a replay strategy for incremental learning, while ABR [25] and our method further incorporate knowledge distillation to implicitly transfer the knowledge learned by the teacher model. This augmented new data directly

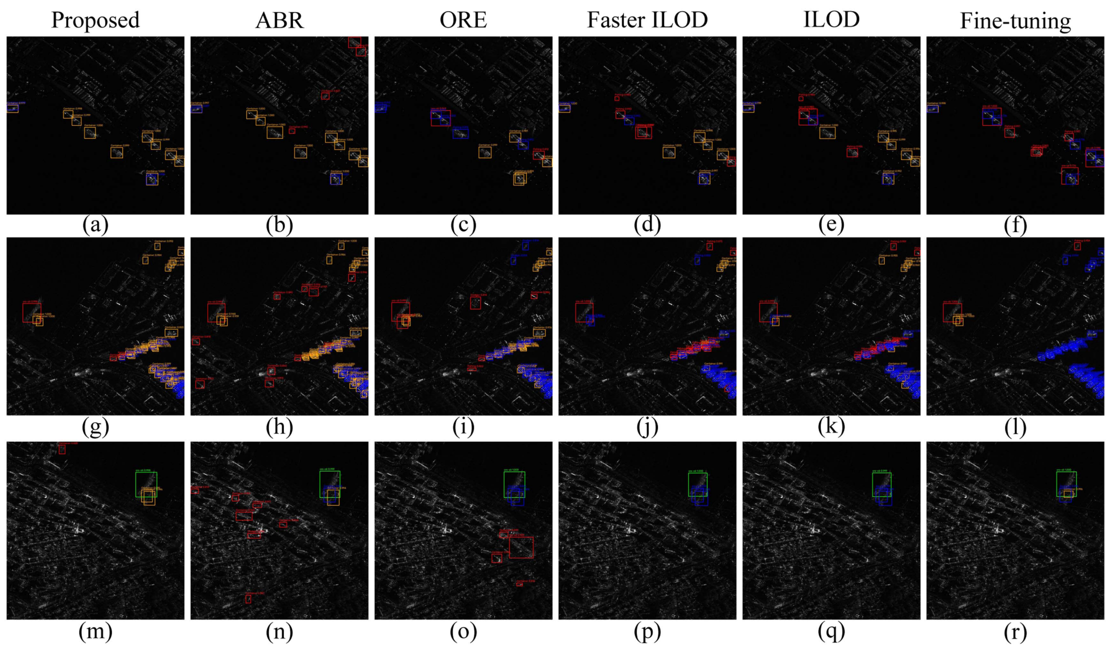

Fig. 7. (a)–(r) Detection and classification results on settings with single-step increments, where the green rectangles denote the new targets correctly classified, the orange rectangles denote the old targets correctly classified, the red rectangles denote false alarms, and the blue rectangles denote missing detections. The content above the predicted bounding box indicates the predicted class label and the classification probability.

include a small number of old targets as well as completely new targets. Therefore, compared to regularization-based incremental methods, replay-based incremental methods achieve significant performance improvement at the cost of certain storage overhead. Compared to ORE [23], ABR [25] achieves higher F1 scores on old targets (60.47% versus 53.21%). Our method incorporates the scene characteristics of ship targets, avoiding context-bias issues when replaying old targets, enhancing the ability of the detector to distinguish between ship targets and clutter. The F1 scores on old targets of our method and the overall F1 scores are 8.14% and 5.69% higher than ABR.

Fig. 7 shows some detection and classification results of different methods on settings with single-step increments on the SRSDD-v1.0 dataset. It can be observed that the fine-tuning method fails to accurately detect old targets, resulting in numerous missing detections. Regularization-based incremental methods ILOD and Faster ILOD can detect a certain number of old targets but still suffer from numerous missing detections. Replay-based methods ABR and ORE tend to generate many false alarms, limiting their detection and classification performance. Our method effectively detects and classifies both new and old targets while retaining low false alarms, achieving better detection and classification performance.

2) *Two-Step Increments:* We conduct experiments on the settings with two-step increments on the SRSDD-v1.0 dataset. The catastrophic forgetting and context-bias issues are more crucial on the longer incremental settings, as shown in Table III. Our method consistently outperforms other methods across different incremental tasks, showing 2.38% improvement over the ABR method in Task 1 and 1.98% improvement in Task 2. In addition, it is interesting to observe that in the second incremental training stage, methods that use replay strategies (ORE, ABR, and our method) show a smaller decline in the F1-scores on old targets compared to methods that do not use replay strategies (ILOD and Faster ILOD). After completing incremental Task 2, F1 scores on old targets of the proposed method decreased by 2.46% compared to the teacher model, whereas Faster ILOD shows a decline of 7.27%. We argue that directly replaying old targets is more effective in mitigating catastrophic forgetting than implicit knowledge distillation.

Fig. 8 shows some detection and classification results of different methods on settings with two-step increments on the SRSDD-v1.0 dataset. We observe that replay-based methods can detect and classify more targets compared to regularization-based methods, although they also tend to produce more false alarms. In addition, our method maintains lowest false alarms compared to other methods.

E. Model Analysis

1) *Analysis of the Context-Robust ER:* To analyze the context-robust ER method, we implement ablation experiments on the SRSDD-v1.0 dataset on the settings with single-step increments, as shown in Table IV. Context-robust ER mainly consists of two parts: 1) local context-aware based on new targets (represented as Rob.) and 2) ER (represented as Aug.). Therefore, our main focus is on analyzing the impact of Rob. and Aug. on the performance of the model. One without Rob. implies training the detector using a traditional ER strategy. We maintain consistency by controlling other variables in the experiments.

It can be observed by comparing the first and second rows in Table IV that with the introduction of Aug., the F1 score on old targets shows significant improvement.

TABLE III
F1 Scores on Settings With Two-Step Increments on the SRSDD-V1.0 Dataset

F1 (%) / Method	Two-step increments (2+2+2) Task D_1:2+2 Old	New	Overall	Task D_2:(2+2)+2 Old	New	Overall
Fine-tuning	23.17±1.78	37.48±0.62	30.32±1.20	20.95±1.27	59.92±1.68	32.89±0.36
ILOD [17]	48.27±1.31	41.21±2.05	44.73±1.45	36.80±1.04	63.74±1.20	45.69±0.49
F-ILOD [19]	49.96±1.43	41.98±1.69	45.97±1.30	38.70±1.71	61.12±1.99	46.17±0.66
ORE* [23]	50.36±1.59	42.42±0.55	46.39±1.07	45.58±2.09	62.04±1.90	51.06±1.69
ABR* [25]	55.77±1.83	**44.65**±0.98	50.21±1.40	47.67±1.48	62.39±3.55	52.58±0.86
Proposed*	**61.48**±1.53	43.70±1.15	**52.59**±0.57	**50.13**±0.53	**63.43**±2.32	**54.56**±0.75
Upper bound	64.82±0.36	48.42±0.17	56.62±0.25	75.58±0.22	66.02±0.19	70.52±0.16

We present the mean and standard deviation after three random trials. Bold indicates the best results in each column, while underlined denotes the second best. Methods using exemplars are marked with an asterisk *.

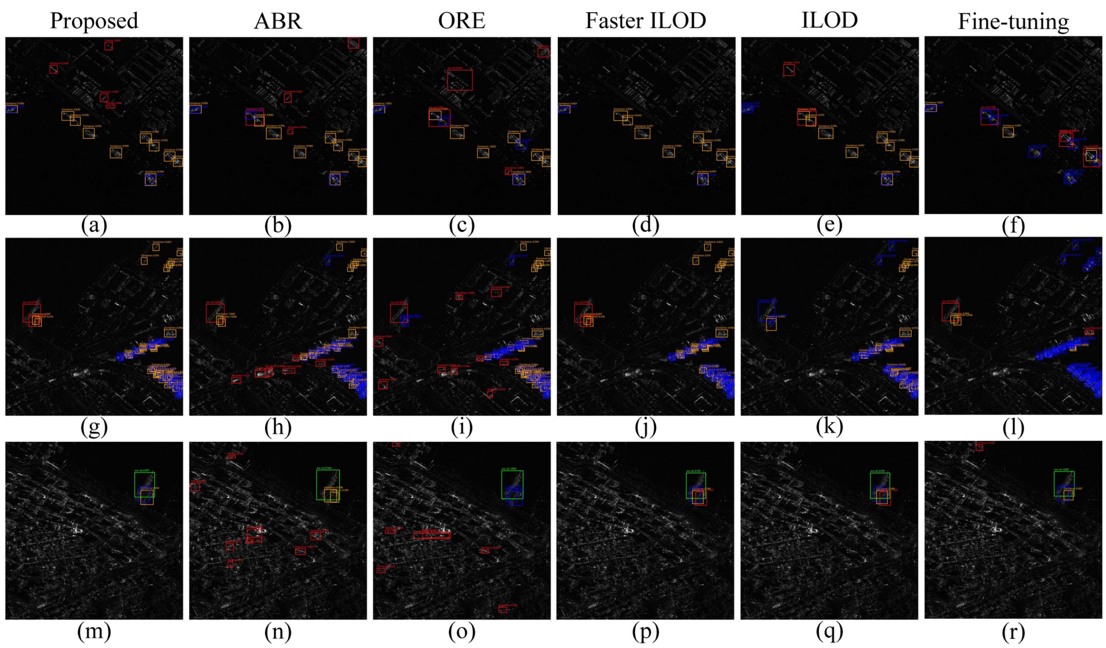

Fig. 8. Detection and classification results on settings with two-step increments, where the green rectangles denote the new targets correctly classified, the orange rectangles denote the old targets correctly classified, the red rectangles denote false alarms, and the blue rectangles denote missing detections. The content above the predicted bounding box indicates the predicted class label and the classification probability.

Meanwhile, the F1 score on new targets also shows a certain degree of improvement (60.54% versus 53.21%). Upon the introduction of Rob., the F1 score on old targets improves by 7.33% and the overall F1 scores increase by 3.31%. The performance improvements shown in the third row further validate that context-robust ER method can effectively retain the knowledge of discriminating old targets from clutter in the incremental training stage.

Fig. 9 shows the difference between traditional ER and proposed context-robust ER methods in two nearshore scenes on the settings with single-step increments. It can be seen intuitively that in these two images, inland regions occupy more than half of the entire image. Compared to the traditional replay method (the second column), the proposed context-robust ER method determines the replay position of old targets based on the local context of new targets. Fig. 10 shows the training loss and test F1 score curves with iterations in the incremental training stage for both the traditional and proposed replay methods. We observe that our replay method achieves a significantly higher old-class F1 score than the traditional replay method (61.19% versus 51.91%) under conditions where both methods converge in training. We attribute this to our contributions in mitigating the context-bias issues induced by the traditional replay

TABLE IV
Ablation Experiments on Settings With Single-Step Increments

Inds	Context-robust Exemplar Replay		Multi-granularity Knowledge Distillation			F1 (%)		
	Local con. Rob.	Exem. Aug.	Sce. $\mathcal{L}_{\text{scene}}$	Ins. $\mathcal{L}_{\text{instance}}$	Res. $\mathcal{L}_{\text{response}}$	Old	New	Overall
①	-	-	-	-	-	14.76±1.77	47.39±0.71	31.07±0.95
②	-	✔	-	-	-	53.21±2.28	57.21±2.80	55.71±0.76
③	✔	✔	-	-	-	<u>60.54</u>±0.82	<u>57.50</u>±1.26	<u>59.02</u>±0.91
④	-	-	-	-	✔	41.59±0.11	54.03±0.08	47.81±0.08
⑤	-	-	✔	-	✔	47.44±0.47	54.35±0.44	50.89±0.39
⑥	-	-	-	✔	✔	44.97±0.17	53.07±0.51	48.97±0.38
⑦	-	-	✔	✔	✔	48.38±0.34	55.26±0.36	51.82±0.30
⑧	✔	✔	✔	✔	✔	**68.61**±0.66	**60.51**±0.62	**64.56**±0.63

We present the mean and standard deviation after three random trials. Bold indicates the best results in each column, while underlined denotes the second best.

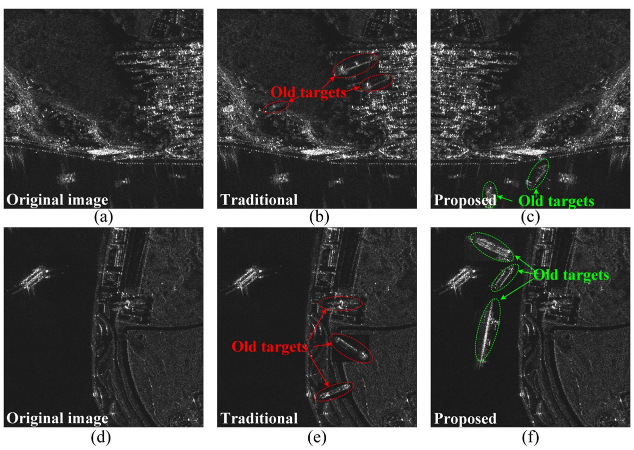

Fig. 9. Comparison of different ER methods in two nearshore scenes at the 10 000 iterations. In both original training images (a) and (d), the inland area occupies more than half of the original image pixels. The traditional replay method is highly likely to place old targets inland as shown in (b) and (e), whereas the proposed context-robust ER method ensures that old targets are placed on the sea surface regions as shown in (e) and (f).

method, thereby reducing the likelihood of the detector missing old targets on the sea surface and falsely detecting background clutter on land.

2) *Analysis of the Multigranularity KD:* To analyze the multigranularity KD method, we implement ablation experiments on the SRSDD-v1.0 dataset on settings with single-step increments, as shown in Table IV. Multigranularity KD mainly consists of three parts: 1) scene-level feature distillation loss (represented as $\mathcal{L}_{\text{scene}}$); 2) instance-level feature distillation loss (represented as $\mathcal{L}_{\text{instance}}$); and 3) top-level response distillation loss (represented as $\mathcal{L}_{\text{response}}$). Therefore, our main focus is on analyzing the impact of $\mathcal{L}_{\text{scene}}$, $\mathcal{L}_{\text{instance}}$, and $\mathcal{L}_{\text{response}}$ on the performance of the

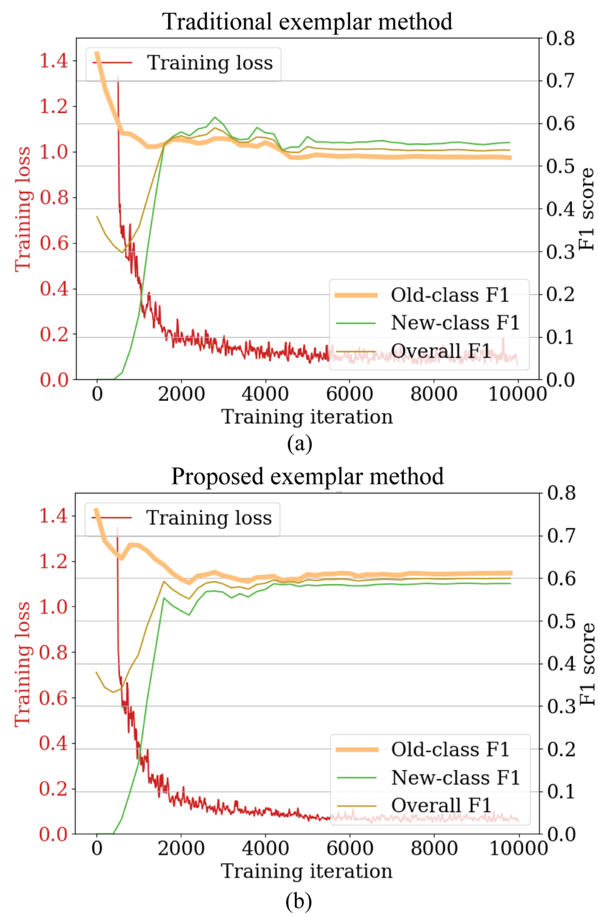

Fig. 10. Training loss and test F1 score curves of the (a) traditional replay method and (b) proposed replay method with respect to iterations in the incremental training stage. Under convergence conditions, the old-class F1 score of the proposed replay method is approximately 10% higher than that of the traditional replay method.

TABLE V
Analysis Experiments on Training Parameter λ Under the Settings With Single-Step Increments

Performance \ λ	0.0	0.5	1.0	1.5	2.0
Old	65.60 ±0.82	67.53 ±0.44	**68.61** ±0.66	<u>67.60</u> ±0.62	64.49 ±0.30
New	60.28 ±1.26	58.93 ±1.11	**60.51** ±0.62	<u>60.34</u> ±0.16	55.86 ±0.26
Overall	62.94 ±0.91	63.23 ±0.99	**64.56** ±0.63	<u>63.97</u> ±0.27	60.17 ±0.28

We present the mean and standard deviation after three random trials. Bold indicates the best results in each column, while underlined denotes the second best.

TABLE VI
Analysis Experiments on Training Parameter q Under the Settings With Single-Step Increments

Performance \ q	2.5	5	10	20	40
Old	63.87 ±1.72	64.65 ±1.37	68.61 ±0.66	<u>69.99</u> ±1.03	**71.10** ±0.96
New	58.50 ±0.98	61.55 ±0.82	60.51 ±0.62	**62.63** ±0.46	<u>61.60</u> ±0.38
Overall	61.18 ±1.03	63.11 ±0.83	64.56 ±0.63	<u>66.31</u> ±0.53	**66.35** ±0.45

We present the mean and standard deviation after three random trials. Bold indicates the best results in each column, while underlined denotes the second best.

TABLE VII
Analysis Experiments on Training Parameter r Under the Settings With Single-Step Increments

Performance \ r	25	50	75	100	125
Old	62.75 ±1.01	**68.61** ±0.66	<u>68.38</u> ±1.23	66.95 ±1.37	62.97 ±0.94
New	59.33 ±0.37	60.51 ±0.62	<u>60.69</u> ±0.67	**60.83** ±0.78	58.39 ±0.65
Overall	61.04 ±0.87	**64.56** ±0.63	<u>64.53</u> ±0.91	63.89 ±1.07	60.68 ±0.69

We present the mean and standard deviation after three random trials. Bold indicates the best results in each column, while underlined denotes the second best.

model. We maintain consistency by controlling other variables during the experiments.

From the fourth row of Table IV, it can be observed that adding $\mathcal{L}_{\text{response}}$ mitigates the forgetting (F1 score on old targets reaches 41.59%), thereby enabling the student model to correctly detect and classify the old targets. We show that our $\mathcal{L}_{\text{scene}}$ and $\mathcal{L}_{\text{instance}}$ improve over $\mathcal{L}_{\text{response}}$ due to better exploitation of target localization information and feature information of the RoIs, resulting in higher F1 scores (The F1 scores on old targets increase by 5.85% and 3.38%, respectively, while overall F1 scores increase by 3.08% and 1.16%, respectively, as shown in the fifth and sixth rows). From the results in the seventh row, it can be observed that the proposed multigranularity KD achieves the optimal F1 score. Compared to only transferring top-level response knowledge, F1 scores on old targets increase by 6.79%, on new targets by 1.23%, and the overall F1 score improved by 4.01%.

3) *Effect of the Training Parameters:* In the incremental training stage, the proposed method primarily involves: 1) the parameter λ that controls the weight of $\mathcal{L}_{\text{distill}}$; 2) the parameter q that controls the replay proportion for each old target class; and 3) the parameter r that controls the radius of the regions for replaying old targets.

In Table V, we explore the influence of the training parameter λ. When $\lambda = 0$, the student model undergoes fully supervised training solely with the context-robust ER method. Since the context-robust ER method replays stored old targets on the new data, the F1 scores of the old targets can still reach 65.60%. When $\lambda = 1.0$, the overall F1 scores of the student model reach the optimal values. However, as λ continues to increase, the performance of the student model declines. This decline may result from an overemphasis on knowledge transfer from the teacher model, which detracts from the student model to directly learn from the augmented new data.

In Table VI, we explore the influence of the training parameter q. It is obvious that as the proportion of stored old targets q increases, the growth rate of F1 scores tends to be stable. Following the majority of replay-based method settings, we store only a small number of old targets ($q = 10$) and replay them in the incremental training stage.

In Table VII, we explore the influence of the training parameter r. When the value of r is 50, 75, or 100, the overall F1 scores is approximately 67%, with the optimal performance observed at $r = 50$. However, when r is either too small or too large, the F1 scores decrease. This is likely because too few local contexts result in insufficient information, while too many can introduce noise, both of which can affect the effectiveness of the local context-aware technique.

V. CONCLUSION

This article proposes an incremental learning method for SAR ship detection and classification. We identify the context-bias issues present in existing replay strategies and propose a context-robust ER method that fully considers the inherent scene characteristics of ship targets. This method ensures that the detector accurately understands the context between old targets and clutter. Meanwhile, a multigranularity KD method is proposed to progressively transfer the knowledge learned by the old detector, ensuring both the performance of the distillation process. Experiments on the SRSDD-v1.0 dataset indicate that our approach offers a significant performance advantage compared to other incremental object detection and classification methods. In addition, we adopt Faster R-CNN as the baseline, but the proposed context-robust ER method can be easily extended to one-stage detectors. We hope the perspectives presented in this article will inspire further research.

A major challenge in incremental learning is acquiring training samples for new classes. While manually collecting labeled data is a common approach, a more effective strategy is enabling the recognition system to detect new classes autonomously. Open set recognition (OSR) [37] enables the model to recognize old classes while identifying unknown new classes. The integration of OSR with incremental learning allows the model to identify new classes and further learn them without retraining [38], [39]. In future work, we will build on prior OSR research [37] and further explore its applications for incremental learning in SAR image analysis.

Incremental learning approaches require a large amount of new data, which poses challenges for SAR ship detection and classification, as annotating extensive SAR datasets is both costly and labor intensive. Our future work will focus on incremental learning in few-shot or semisupervised settings.

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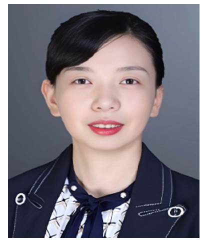

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